

TO GOVERNMENT OF NEPAL
Ministry of Agriculture and Livestock Development

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NEPAL AGRICULTURAL TRANSFORMATION MASTER ROADMAP 2025–2030

A Structural Reform Blueprint for Lawmakers & Agriculture Ministers

NPR 340 Bn Five-Year Investment	10 Root CausesAddressed Directly	3× Yield Target By 2030
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April 2026 | Kathmandu, Nepal

EXECUTIVE SUMMARY

Nepal's agricultural sector contributes approximately 24% of GDP and employs over 60% of the population, yet it operates at roughly half its productive potential. The current annual Ministry of Agriculture and Livestock Development (MoALD) budget stands at NPR 57.48 billion (FY 2025/26) — of which nearly NPR 27.95 billion (49%) is consumed by fertiliser subsidies alone. This structural budget imbalance, combined with ten deeply interconnected root-cause failures, has kept agriculture in a vicious cycle of stagnation for decades.

This roadmap does not propose more of the same. It proposes a fundamental restructuring of how Nepal's agricultural budget is allocated, governed, and executed — shifting from subsidy dependency to capital investment, from top-down extension to farmer-centred services, and from fragmented governance to accountable delivery.

The Ten Root Causes This Roadmap Directly Addresses

#	Root Cause	Primary Impact	5-Year Budget Allocation
1	Structural Land Fragmentation	Mechanisation impossible; no economies of scale	NPR 18 Bn
2	Irrigation Crisis (only 36% coverage)	Yield ceiling at 50% potential; climate vulnerability	NPR 85 Bn
3	Input Underutilisation (fertiliser, seeds)	2–4× yield gap versus South Asian neighbours	NPR 42 Bn
4	Climate Change Acceleration	Unpredictable monsoons; temperature yield losses	NPR 14 Bn
5	Labour Exodus (730,000/year)	32% of hill land lies fallow	NPR 22 Bn
6	Market System Failure	Farmers receive only 30–40% of retail price	NPR 28 Bn
7	Credit & Finance Starvation	Pre-harvest capital inaccessible for small farmers	NPR 35 Bn
8	Human Capital / Extension Collapse	1 technician per 1,500 farmers (4× gap)	NPR 18 Bn
9	Soil Degradation	Long-term productive capacity silently declining	NPR 12 Bn
10	Policy & Governance Failure	Policies exist; implementation doesn't	NPR 66 Bn (governance reform)

TOTAL PROPOSED FIVE-YEAR INVESTMENT: NPR 340 BILLION

(Average NPR 68 billion/year — a 18% increase over current allocation, with radical reallocation away from unproductive subsidies toward capital investment)

Expected Outcomes by 2030

Indicator	Baseline (2025)	Target (2030)
Irrigated land (% of agricultural area)	36%	65%
Average paddy yield (tonnes/ha)	3.1	5.5
Fertiliser application (kg/ha)	67.4	140+
Farmers with formal credit access	6–7%	35%
Post-harvest losses (fruits/vegetables)	30–40%	10–15%
Extension ratio (farmers per technician)	1,500:1	500:1
Agricultural exports (NPR billion/yr)	60	180+
Fallow land in hills (%)	32%	12%
Agricultural contribution to GDP growth	~1%/yr	4–5%/yr

PART I: BUDGET REALITY AND STRUCTURAL REFORM

1.1 Current Budget: What Is Wrong

Nepal's agriculture budget has flatlined in real terms. The FY 2025/26 allocation of NPR 57.48 billion represents a mere 0.34% increase over FY 2024/25 — far below inflation. More critically, the internal allocation is deeply dysfunctional:

Budget Item	FY 2024/25 Allocation	% of Total	Assessment
Fertiliser subsidies (single item)	NPR 27.95 Bn	48.8%	Inefficient — reaches only 30–40% of farmers
PM Agriculture Modernisation Project	NPR 2.98 Bn	5.2%	Too small; dispersed across 4,850 ha only
Sub-federal levels (provinces + local)	NPR 4.38 Bn	7.65%	Critically underfunded for devolved mandate
Agricultural research (NARC)	~NPR 3.5 Bn	6.1%	Chronically understaffed; research-extension gap
Irrigation (under MoALD portion)	~NPR 3.0 Bn	5.2%	Insufficient; large irrigation under MoEWRI
Extension services	~NPR 2.5 Bn	4.4%	Cannot serve 3.4 million farm households
All other programmes	~NPR 12.9 Bn	22.5%	Fragmented across 200+ line items

The core problem: Nearly half the agriculture budget creates no lasting productive capacity. Subsidies suppress prices, distort markets, and crowd out capital investment. The proposed roadmap shifts this fundamentally.

1.2 Proposed Budget Reallocation Framework (5-Year)

The roadmap proposes maintaining total spending at NPR 65–75 billion/year (achievable within the declared Agriculture Investment Decade 2081–2091), but with radical internal reallocation:

Category	Current Share	Proposed Share	Rationale
Fertiliser subsidies	49%	20% (targeted)	Replace blanket with smart targeted subsidy
Irrigation infrastructure	5%	20%	Single highest ROI investment available
Credit guarantee & finance	1%	10%	Unlock private bank lending to farmers
Market infrastructure & value chains	2%	10%	Cold storage, roads, aggregation centres
Extension & human capital	4%	10%	Double technician workforce; digital tools
Research & climate adaptation	6%	8%	Translate research to farm-level practice
Land consolidation incentives	0%	6%	Cooperative land pooling incentives
Sub-federal agricultural grants	8%	16%	Restore devolved mandate funding

PART II: THE FIVE-YEAR REFORM PILLARS

PILLAR 1: SOLVING LAND FRAGMENTATION | NPR 18 BILLION

Problem Summary

Average holding size of 0.6 hectares — below every viable mechanisation or commercialisation threshold. Inheritance law, absent land markets, and failed cooperatives have created mathematical impossibility of scale. 50% of land lacks documentation. Hill regions have 32% fallow land due to migration.

Solutions

Solution 1A: Voluntary Land Pooling Programme (LPP)

Do not attempt forced consolidation (politically impossible). Instead, create overwhelming financial incentives for voluntary cooperative land pooling.

- Government provides a capital grant of NPR 80,000 per hectare to any cooperative that pools at least 10 contiguous hectares under a single operational management unit
- Cooperatives that pool 25+ hectares receive subsidised access to tractors, combine harvesters, and irrigation equipment through a machinery leasing scheme
- Pooled land retains individual ownership — farmers lease their land to the cooperative for a minimum 10-year term and receive annual lease payment plus a share of profits
- Tax exemption on all income from pooled cooperative farms for the first 7 years
- Priority government procurement for food grains from pooled farms at guaranteed floor price

BUDGET	LPP capital grants: NPR 8 Bn Machinery procurement for leasing scheme: NPR 6 Bn Programme administration: NPR 1 Bn Total: NPR 15 Bn over 5 years
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Solution 1B: Land Documentation and Titling Drive

50% of agricultural land lacks formal documentation. This blocks credit access, land sales, and cooperative formation.

- Launch a 3-year National Land Documentation Mission: mobilise land survey teams across all 753 local governments
- Issue digital land certificates (integrated with existing LRIMS) with biometric owner verification
- Offer free documentation for holdings under 1 hectare; nominal fee for larger holdings
- Resolve boundary disputes through local land arbitration panels before documentation (avoid court backlogs)
- Link documented land to the Agricultural Credit Registry — enabling formal bank lending

BUDGET	Land documentation mission: NPR 2 Bn Digital LRIMS upgrades: NPR 0.5 Bn Local arbitration panels: NPR 0.5 Bn Total: NPR 3 Bn
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Solution 1C: Inheritance Law Amendment (Legislative Action Required)

This is the only solution requiring Parliament. The current mandatory equal-share inheritance law mathematically guarantees continued fragmentation.

- Amend the Muluki Dewani Samhita to allow family land trusts: family members inherit equal shares in a trust, but the land is not physically subdivided below 0.5 hectares
- Family Land Trust mechanism: one designated member operates the land; others receive annual trust income; any member can sell their trust share to another family member or exit through a buyout mechanism
- This preserves inheritance rights without physical subdivision
- Complementary: establish a government-backed Land Bank that purchases highly fragmented plots below 0.3 ha, consolidates them, and leases to agricultural cooperatives

BUDGET	Land Bank capitalisation: NPR 0 Bn Year 1–2 (requires NPR 5 Bn seed capital from future budgets — start legislation Years 1–2; fund Year 3+)
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Year-by-Year Milestones — Pillar 1

Year	Land Documentation	Land Pooling	Legal Reform
Year 1 (2025/26)	Survey teams deployed in all 77 districts	LPP scheme designed & announced; 100 pilot cooperatives	Inheritance law amendment drafted
Year 2 (2026/27)	50% of undocumented land titled	500 cooperatives, 5,000 ha pooled	Amendment in Parliament
Year 3 (2027/28)	80% documentation achieved	2,000 cooperatives, 20,000 ha pooled	Law enacted; Land Bank established
Year 4 (2028/29)	Full documentation in Terai	50,000 ha under operational pooling	Land Bank purchasing fragmented plots
Year 5 (2029/30)	National land registry complete	100,000 ha pooled nationally	Trust mechanism operational

PILLAR 2: SOLVING THE IRRIGATION CRISIS | NPR 85 BILLION

Problem Summary

Only 36% of agricultural land has year-round irrigation despite Nepal receiving 1,600 mm annual rainfall. 80% of rain falls in 4 monsoon months. Canal efficiency is critically low. Farmer-managed irrigation systems deteriorate without funding. Groundwater governance is split across three government tiers with no one accountable. This is Nepal's single highest-return investment.

Solutions

Solution 2A: National Small-Scale Irrigation Acceleration Programme

Prioritise small and medium irrigation schemes (5–500 ha command area) over mega projects. These have faster construction timelines, lower cost per hectare, and can be managed locally.

- Target: add irrigation coverage to 500,000 additional hectares over 5 years (raising coverage from 36% to 65%)
- Government funds 80% of construction cost for schemes serving smallholders (< 1 ha average); farmers contribute 20% in-kind labour
- All new schemes must incorporate lined canals (canal efficiency target: minimum 50% — up from current sub-30% level)
- Mandatory: each scheme must have a legally registered Water User Association (WUA) with an elected committee, fee-collection authority, and maintenance endowment fund before construction begins
- WUAs receive an annual government maintenance grant of NPR 3,000/ha for the first 5 years, tapering to NPR 1,500/ha thereafter — solving the 'free rider' maintenance collapse problem

BUDGET	500,000 ha × avg NPR 80,000/ha construction cost × 80% govt share = NPR 32 Bn WUA maintenance grants (5 yr): NPR 7 Bn Canal rehabilitation (existing 64% coverage): NPR 8 Bn Total: NPR 47 Bn
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Solution 2B: Monsoon Water Harvesting and Dry-Season Storage

The fundamental hydrological problem — 80% of rain in 4 months — requires storage infrastructure. Small-scale farm ponds and micro-reservoirs are the most economically viable solution.

- Community farm pond programme: government provides NPR 150,000 grant per farm pond (capacity: 500–2,000 m³) serving 2–5 farmers in cluster; target 200,000 ponds over 5 years
- Medium check dams (capacity 50,000–500,000 m³): construct 500 check dams in hill watersheds; government funds 100% — these are public goods with multiple benefits (irrigation, groundwater recharge, flood control)
- Rooftop rainwater harvesting: mandatory in all new agricultural processing facilities; 50% subsidy for farm-level plastic pond liners (critical for hill farmers who cannot dig deep ponds)
- Integrate water storage siting with climate projections — design for 2040 hydrological patterns, not 2010 averages

BUDGET	200,000 farm ponds × NPR 150,000: NPR 30 Bn 500 check dams × avg NPR 10 Mn: NPR 5 Bn Rooftop/liner subsidies: NPR 1 Bn Total: NPR 36 Bn over 5 years
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Solution 2C: Groundwater Governance Reform

Only 4.7% of available groundwater is utilised. No government tier manages it effectively after federalisation. This requires an institutional fix, not just capital.

- Enact a National Groundwater Management Act: assign clear authority — local governments regulate extraction; provinces maintain aquifer data; federal government sets standards
- Establish Provincial Groundwater Monitoring Networks: 500 observation wells per province to track water table levels seasonally
- Subsidised shallow tube wells for smallholders (< 1 ha): 75% capital subsidy, capped at NPR 120,000 per well; conditional on WUA membership and extraction metering
- Conjunctive use training programme: extension workers trained specifically to advise farmers on combining surface water (monsoon) with groundwater (dry season) — this alone can add a second crop cycle
- Prohibit deepening of wells beyond 60m without provincial licence in water-stressed zones

BUDGET	Groundwater Act & monitoring networks: NPR 1 Bn 50,000 subsidised shallow tube wells: NPR 1 Bn (balance farmer-funded; NPR 0.5 Mn public subsidy each) Total: NPR 2 Bn
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Year-by-Year Milestones — Pillar 2

Year	Small-Scale Irrigation	Water Storage	Groundwater
Year 1	100,000 ha new coverage; WUA framework enacted; canal rehabilitation begins	50,000 farm ponds constructed; 100 check dams	Groundwater Act drafted; 200 monitoring wells
Year 2	200,000 ha total; 500 WUAs operational	100,000 farm ponds; 250 check dams	Act enacted; 500 monitoring wells
Year 3	320,000 ha total; canal efficiency 40%+ on new schemes	150,000 ponds; 400 check dams; seasonal water security for 500,000 farmers	50,000 tube wells; conjunctive use programme launched
Year 4	430,000 ha; WUA maintenance funding stabilised	180,000 ponds; 500 check dams complete	Groundwater maps completed for all provinces
Year 5	500,000 ha new coverage; national coverage 65%	200,000 ponds; water storage adds 8-month security for 1 Mn farmers	Full groundwater governance operational

PILLAR 3: SOLVING INPUT UNDERUTILISATION | NPR 42 BILLION

Problem Summary

Nepal uses 67.4 kg/ha of fertiliser versus 163.5 kg/ha in India and 279 kg/ha in Bangladesh. 100% of chemical fertiliser is imported. The MOU with India expired March 31, 2026, leaving Nepal's paddy season stocks 27% below requirement. Seed systems have collapsed — most farmers use degraded saved seeds. Extension advice is generic and misaligned to local conditions. These three failures together create Nepal's 2–4× yield gap.

Solutions

Solution 3A: Fertiliser Supply Security and Smart Subsidy Reform

The current NPR 27.95 Bn blanket fertiliser subsidy is Nepal's largest single agricultural expenditure. It reaches only 30–40% of farmers. It must be restructured, not simply cut.

- Immediate action (Year 1): negotiate emergency fertiliser supply agreements with at least THREE sources — India, Bangladesh (via India transit), China (via Tibet border), and explore Middle East DAP suppliers — to end single-source dependency
- Establish a National Fertiliser Strategic Reserve: minimum 90-day buffer stock (approximately 65,000 tonnes) maintained at provincial distribution centres; funded by a NPR 2 Bn reserve fund
- Replace blanket subsidy with a Smart Farmer Card system: every registered farmer receives a digital card with a annual fertiliser entitlement based on their documented land area (1 hectare = entitlement for 140 kg at subsidised price)
- Smart cards prevent elite capture: subsidy is quantity-capped per farmer, cannot be transferred, and requires purchase from licensed retailers only
- Phase fertiliser subsidy down from NPR 27.95 Bn to NPR 14 Bn over 5 years — reinvesting the NPR 14 Bn savings into irrigation, extension, and credit guarantee
- Invest NPR 3 Bn in domestic production feasibility: study and pilot organic/bio-fertiliser production from Nepal's abundant livestock waste — long-term path to import independence

BUDGET

Strategic reserve establishment: NPR 3 Bn | Smart card digital system: NPR 1 Bn | Subsidy reform operational costs: NPR 1 Bn | Bio-fertiliser pilot plants: NPR 3 Bn | Fertiliser subsidy (reformed, 5-yr total): NPR 70 Bn (reduced from current ~NPR 140 Bn trajectory) | Net saving: NPR 70 Bn redirected to other pillars

Solution 3B: National Seed System Overhaul

Most farmers use degraded saved seeds yielding 40–60% below potential of improved varieties. The public seed system has collapsed. Private seed markets are emerging but farmers cannot access quality seeds due to cost and information gaps.

- Establish 7 Provincial Seed Production Farms (one per province): government-owned, professionally managed, producing breeder and foundation seed for locally appropriate varieties; target 10,000 tonnes of quality seed/year by Year 3

- Seed voucher programme: all farmers in the LPP (Pillar 1) and irrigation beneficiaries (Pillar 2) receive a one-time seed voucher worth NPR 5,000 for transition to improved varieties — linked to extension training on proper use
- Climate-adaptive variety fast-tracking: NARC mandate to release minimum 10 new climate-adapted varieties per year (drought-tolerant rice, heat-tolerant wheat, early-maturing maize); current rate is 2–3/year
- Community Seed Banks: fund 1,000 community seed banks at local government level; these preserve local diversity while providing access to improved varieties; each costs NPR 200,000 to establish and equip
- Abolish seed import duties: allow private seed companies to import hybrid seeds tax-free — increase competition and reduce prices to farmers

BUDGET

7 Provincial Seed Farms: NPR 5 Bn | Seed voucher programme: NPR 3 Bn | Community seed banks: NPR 0.2 Bn | NARC climate variety programme: NPR 2 Bn | Total: NPR 10.2 Bn

Year-by-Year Milestones — Pillar 3

Year	Fertiliser Security	Seed System	Target Yield Impact
Year 1	3-source supply agreements signed; reserve fund established	2 Provincial Seed Farms operational; seed vouchers piloted in 10 districts	Yield gap reduction: 10%
Year 2	Smart Farmer Card system launched; strategic reserve stocked	5 Provincial Seed Farms; 10 climate varieties released; 500 community seed banks	Yield gap reduction: 20%
Year 3	Smart card coverage 70% of farmers; subsidy down to NPR 20 Bn	All 7 Seed Farms; 1,000 community seed banks; 500,000 farmers on improved varieties	Average yield up 30%
Year 4	Subsidy NPR 16 Bn; domestic bio-fertiliser pilot scaling	1 million farmers on improved varieties; private seed market competitive	Average yield up 45%
Year 5	Subsidy NPR 14 Bn; 90-day buffer stock permanent	2 million farmers on improved seeds; domestic seed self-sufficiency 40%	Fertiliser use 140 kg/ha; avg yield up 60%+

PILLAR 4: MARKET SYSTEM TRANSFORMATION | NPR 28 BILLION

Problem Summary

Farmers receive only 30–40% of the retail price of their produce. Value chains are 7–9 stages long. Post-harvest losses are 30–40% for perishables. Nepal is the world's largest cardamom producer and a leading ginger producer, yet agricultural exports are only NPR 60 billion/year. No government department has a dedicated value chain mandate. Infrastructure isolation forces distress sales.

Solutions

Solution 4A: Agricultural Market Infrastructure Network

- Construct 77 District Aggregation Centres (one per district): each centre has cold storage (500 tonne capacity), grading and sorting equipment, truck docking, and a price information terminal; total cost NPR 50 Mn per centre
- Construct 7 Provincial Agro-Processing Hubs (one per province): large-scale processing capacity for major commodities (ginger drying, cardamom processing, tea processing, pulse milling); government builds shell infrastructure, private operators lease at commercial rates
- Last-mile connectivity: earmark NPR 5 Bn for agricultural feeder roads connecting villages to market centres — prioritise roads that unlock currently fallow hill land
- Cold chain: 50% capital subsidy for private cold storage investment below NPR 50 Mn — government backs but does not own; target 200 new private cold stores in 5 years

BUDGET	77 District Aggregation Centres: NPR 3.85 Bn 7 Provincial Processing Hubs: NPR 5 Bn Feeder roads: NPR 5 Bn Cold chain subsidies: NPR 3 Bn Total: NPR 16.85 Bn
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Solution 4B: Export Value Chain Acceleration

Nepal has world-class comparative advantage in cardamom, ginger, tea, honey, herbs, and high-altitude vegetables. These are underexploited due to certification gaps, supply inconsistency, and poor branding.

- Establish a dedicated Agricultural Export Promotion Board (separate from Trade and Export Promotion Centre): focused exclusively on agricultural commodities; mandate: organic certification support, buyer connections, trade mission participation
- Collective certification: government funds group organic and GlobalG.A.P. certification for cooperatives of 50+ farmers — individual certification is too costly; groups spread the cost
- Nepal Agricultural Brand: launch a single government-backed 'Himalayan Origin' brand for premium Nepali agricultural products — reduces buyer search costs and allows premium pricing
- Processing investment incentives: 7-year tax holiday + 50% import duty exemption on machinery for agro-processing investments above NPR 50 Mn that commit to purchasing from Nepali farmers

- Ginger and cardamom value chain deepening: Nepal earns 8–10% of final consumer price; target 25% through domestic processing (essential oils, extracts, packaged spices)

BUDGET	Export Promotion Board: NPR 2 Bn (5-yr operational) Collective certification support: NPR 1 Bn Himalayan Origin brand development: NPR 0.5 Bn Processing investment incentives (tax revenue forgone, not expenditure): estimated NPR 3 Bn Total direct expenditure: NPR 3.5 Bn
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Solution 4C: Digital Market Information and Direct Trade Platform

- National Agricultural Price Information System: real-time wholesale prices broadcast via SMS and a free app to all registered farmers — eliminates information asymmetry that enables middlemen exploitation
- Government-backed digital direct trade platform: connects farmer cooperatives (minimum 20 members) directly to institutional buyers (schools, hospitals, army canteens, food processors) — government is anchor buyer, committing to purchase 15% of public institutional food procurement from registered cooperatives
- Government procurement as market anchor: commit MoE (school meals), MoHP (hospitals), Nepal Army canteen, and Nepal Police canteen to source minimum 20% of food from Nepali farmers through this platform by Year 3

BUDGET	Price information system: NPR 0.8 Bn Digital trade platform: NPR 0.5 Bn Government procurement premium (above market price paid to farmers): NPR 7 Bn over 5 years Total: NPR 8.3 Bn
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PILLAR 5: AGRICULTURAL FINANCE TRANSFORMATION | NPR 35 BILLION

Problem Summary

Only 6.7% of agricultural credit goes to pre-harvest activities — the exact moment capital is most needed. Banks charge 18–24% interest for agricultural loans — double industrial rates. Small farmers lack collateral. ADBL reaches only 5% of the agricultural population. Microfinance rates (18–24%) exceed agricultural profit margins (10–15%). The result: subsistence farming persists because commercialisation requires capital that is structurally inaccessible.

Solutions

Solution 5A: Agricultural Credit Guarantee Fund

The single most powerful financial intervention: de-risking lending to small farmers so private banks will lend at viable rates.

- Establish a NPR 20 Bn Agricultural Credit Guarantee Fund (ACGF): government-capitalised, professionally managed; guarantees 70% of loss on qualifying agricultural loans
- Qualifying criteria: loans to farmers with documented land (Pillar 1); used for pre-harvest inputs, irrigation, mechanisation, or post-harvest storage; maximum NPR 500,000 per borrower
- With 70% guarantee, bank's net risk reduces from 100% to 30% — this justifies interest rate reduction from 18–24% to 10–12%
- ACGF target: mobilise NPR 100 Bn of private bank lending to farmers over 5 years from NPR 20 Bn government capital (5:1 leverage ratio)
- Nepal Rastra Bank directive: all A-class commercial banks must allocate minimum 5% of loan portfolio to agricultural credit (enforce existing but unenforced directive with penalties)

BUDGET

ACGF capitalisation: NPR 20 Bn | ACGF management: NPR 0.5 Bn | Total: NPR 20.5 Bn. Expected leverage: NPR 100 Bn in private bank lending unlocked.

Solution 5B: Seasonal Agricultural Credit Matching Farmer Cash Flow

Fixed monthly repayment schedules are incompatible with agricultural income cycles. This structural mismatch is why farmers default and why banks won't lend.

- Mandate all ACGF-guaranteed loans to use harvest-linked repayment: principal + interest due within 60 days of harvest season (aligned to crop calendar, not monthly schedule)
- NRB creates a standardised 'Agricultural Seasonal Loan' product: 6–10 month term, disbursed before planting, repaid after harvest; banks cannot impose monthly payments on this product
- Warehouse receipt financing: farmers store produce in certified district aggregation centres (Pillar 4), receive a warehouse receipt, and use it as collateral for immediate post-harvest loans — eliminates distress selling; government subsidises warehouse receipt system for first 3 years

- Input supplier credit linkage: government negotiates with major fertiliser and seed suppliers to provide 90-day deferred payment terms to ACGF-registered farmers — reduces the need for cash credit before planting

BUDGET

Warehouse receipt system development: NPR 2 Bn | NRB regulatory framework (technical assistance): NPR 0.3 Bn | Input supplier credit facilitation: NPR 0.2 Bn | Total: NPR 2.5 Bn

Solution 5C: ADBL Reform and Digitalisation

- ADBL branch expansion: open 100 new agricultural service centres in hill and mountain districts where ADBL has zero presence; these can be small digital banking points, not full branches (cost NPR 5 Mn each)
- Digital loan processing: mobile-based loan application for amounts under NPR 100,000 using digital land records (Pillar 1) as collateral verification — reduces transaction cost from NPR 15,000 to NPR 2,000 per loan, making small loans viable
- Group lending model: 10-member farmer groups can take joint loans with collective liability — each member guarantees others; reduces individual risk; proven in Bangladesh (Grameen) and India (SHG model)
- ADBL recapitalisation: inject NPR 12 Bn fresh equity into ADBL to expand lending capacity to hill regions

BUDGET

100 new ADBL service centres: NPR 0.5 Bn | Digital loan system: NPR 0.3 Bn | ADBL recapitalisation: NPR 12 Bn | Total: NPR 12.8 Bn

PILLAR 6: EXTENSION & HUMAN CAPITAL TRANSFORMATION | NPR 18 BILLION

Problem Summary

Nepal has 1 extension technician per 1,500 farmers. The global standard is 1 per 400. Agricultural college curricula are decades old. Of 1,100 agriculture graduates annually, few enter extension or start agricultural enterprises — most migrate. Extension advice is top-down, not farmer-problem driven. Digital extension projects launch but collapse after donor funding ends.

Solutions

Solution 6A: Mass Extension Technician Recruitment and Deployment

- Hire 6,000 new Junior Agricultural Extension Officers (JAEOs) over 3 years: target ratio of 1:500 by Year 4
- JAEOs must be deployed to their home district — local knowledge is essential; prohibit deployment to other provinces for first 5 years
- JAEO salary: NPR 35,000/month + NPR 10,000 field allowance (competitive with private sector to retain talent in rural areas)
- Performance incentives: each JAEO receives a bonus of NPR 50,000/year if average yield in their cluster increases more than 15% in a year — creates accountability
- Mobile extension: each JAEO provided with a motorbike (government-owned) and a tablet with offline-capable farm diagnosis app — all funded

BUDGET	6,000 JAEOs × NPR 540,000/yr salary+allowance × 5 yr (avg 4,000 deployed) = NPR 10.8 Bn Motorbikes + tablets: NPR 1 Bn Training: NPR 0.5 Bn Total: NPR 12.3 Bn
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Solution 6B: Agricultural Curriculum Emergency Reform

- Mandate all 20 agriculture colleges to adopt updated curricula by Year 2: syllabi must include climate-adaptive farming, digital agriculture, agri-business management, and cooperative management — not merely theoretical crop science
- Establish 7 Provincial Agricultural Vocational Training Centres (one per province): 6-month practical programmes for working farmers — not 4-year degrees; target 50,000 farmers trained/year in practical techniques
- Farmer Field Schools: each JAEO runs 2 Farmer Field Schools per year (25 farmers per school, season-long practical learning on real plots); 6,000 JAEOs × 2 schools × 25 farmers = 300,000 farmers trained/year at programme scale
- International exchange: 200 promising JAEOs sent per year to India (ICAR), Bangladesh (BARI), and the Philippines (IRRI) for 3-month attachments — bring back best practices

BUDGET	7 Provincial Vocational Centres: NPR 2 Bn Curriculum reform support: NPR 0.3 Bn International exchange programme: NPR 0.3 Bn Farmer Field Schools (materials): NPR 1 Bn Total: NPR 3.6 Bn
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Solution 6C: Digital Agricultural Advisory System

- Build a National Agricultural Knowledge Platform (NAKP): offline-capable app + SMS service providing: disease and pest identification (photo-based AI diagnosis), crop calendar for each agro-climatic zone, current market prices, weather forecasts, and nearest JAEO contact
- Public-private partnership: NAKP built and maintained by a private operator on a 5-year government contract (government funds development; operator funds maintenance via agri-input advertising — keeping farmer services free)
- Radio and community FM: 50% of Nepali farmers will not use digital tools within 5 years; fund agricultural advisory radio programmes in all 7 provinces covering crop calendars, pest alerts, and market prices

BUDGET	NAKP development: NPR 1 Bn 5-year maintenance contract: NPR 0.5 Bn Radio programme network: NPR 0.5 Bn Total: NPR 2 Bn
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PILLAR 7: CLIMATE RESILIENCE & SOIL HEALTH | NPR 26 BILLION

Climate Resilience — NPR 14 Billion

Solution 7A: Climate-Proof Cropping System Transition

- Mandatory climate vulnerability mapping: all 77 districts must complete agricultural climate risk assessment by Year 1 — identifying which areas face flood risk, drought risk, and heat stress
- Crop zone rebalancing: based on risk maps, provide incentives for farmers in flood-prone areas to shift to flood-tolerant rice varieties; farmers in drought-prone areas receive drought-tolerant variety seed kits at 80% subsidy
- Altitude shifting: in high-altitude zones facing temperature rise, government supports transition to higher-value temperate horticultural crops (apple, walnut, stone fruits) — high margin, climate-appropriate
- Community early warning network: connect all 753 local governments to DHMS (Department of Hydrology and Meteorology) real-time alerts; mandatory evacuation and crop protection protocols before predicted floods
- Agricultural insurance: launch a mandatory weather-indexed crop insurance scheme for all ACGF-credit recipients (Pillar 5); government subsidises 50% of premium for small farmers; insurance prevents credit default cascade when floods destroy crops

BUDGET

Climate risk mapping: NPR 1 Bn | Variety transition subsidies: NPR 3 Bn | Altitude crop transition support: NPR 2 Bn | Insurance premium subsidy: NPR 5 Bn | Early warning network: NPR 1 Bn | Total: NPR 12 Bn

Soil Health — NPR 12 Billion

Solution 7B: National Soil Health Programme

- Soil testing for every farm: deploy mobile soil testing labs (van-based) across all 77 districts; free soil tests for all registered farmers; target 500,000 tests/year; test results integrated into NAKP and JAE0 advice
- Soil health cards: each farmer receives a Soil Health Card every 2 years specifying deficiencies and precise input recommendations — replaces blanket fertiliser advice with precision nutrition management
- Compost and organic matter programme: 50% subsidy on compost pit construction; each local government required to establish a compost aggregation point; target: increase soil organic matter from current 0.5–1% to 2–3% over 10 years
- Conservation agriculture promotion: JAE0s trained in and required to promote minimum tillage, cover cropping, and crop rotation; 10,000 demonstration plots established across Nepal
- Hill terrace rehabilitation: NPR 4 Bn earmarked for stone terrace rehabilitation in hill districts — terraces reduce 70–80% of soil erosion; labour-intensive construction provides rural employment

BUDGET	Mobile soil testing labs (77): NPR 1 Bn Soil health card system: NPR 0.5 Bn Compost programme: NPR 2 Bn Hill terrace rehabilitation: NPR 4 Bn Conservation agriculture demos: NPR 0.5 Bn Total: NPR 8 Bn
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PILLAR 8: GOVERNANCE AND INSTITUTIONAL REFORM | NPR 66 BILLION (SYSTEM REFORM)

Note: The NPR 66 Bn figure represents the reallocation and governance reforms to existing budget flows — primarily the fertiliser subsidy reform and sub-federal funding restoration. Direct governance reform costs are NPR 4 Bn.

Solutions

Solution 8A: Fix Federal-Provincial-Local Agricultural Coordination

- Enact an Agricultural Functions Clarity Act: legally define which agricultural functions belong at which government tier; eliminate the concurrent confusion
- Create a National Agricultural Coordination Council: Finance Minister chairs; Agriculture Minister, seven Provincial Agriculture Ministers, and Federal Farmers' Federation as members; meets quarterly; resolves inter-tier conflicts; this body approves the 5-year programme targets
- Sub-federal funding restoration: increase provincial agriculture allocation from current 2.02% to 8% of MoALD budget; increase local government agriculture allocation from 5.64% to 15%; this funds the local implementation machinery
- Provincial Agricultural Investment Plans: each of the 7 provinces required to submit a 3-year Agricultural Investment Plan to receive their federal conditional grant — creates planning discipline and accountability

BUDGET

Functions Clarity Act (legal drafting, consultation): NPR 0.2 Bn | NACC secretariat: NPR 0.3 Bn | Sub-federal funding increase (absorbed from fertiliser subsidy savings): NPR 8 Bn/yr additional | Direct cost: NPR 0.5 Bn

Solution 8B: Programme Consolidation and Anti-Corruption

- Consolidate 200+ agricultural programme line items into 8 Flagship Programmes (matching this roadmap's 8 pillars): each flagship has a single budget line, single accountable Ministry official, and published quarterly performance reports
- Beneficiary registration: all subsidy recipients must be registered in a central database with biometric ID — prevents duplicate claims and elite capture
- Third-party monitoring: engage independent monitoring agencies to verify physical progress on all capital investments above NPR 50 Mn; results published publicly; penalty clause for contractors
- Anti-corruption unit: establish a dedicated Agricultural Programmes Anti-Corruption Unit under CIAA (Commission for Investigation of Abuse of Authority) with authority to investigate and prosecute subsidy fraud
- Budget execution emergency: Agriculture Ministry capital budget execution rate is below 70% annually. Mandate: any ministry official unable to achieve 85% capital budget execution by Poush (mid-January) faces performance review. Simplify procurement rules for agricultural infrastructure below NPR 50 Mn to 2-stage process.

BUDGET	Programme consolidation (IT, system): NPR 0.5 Bn Beneficiary registry: NPR 1 Bn Third-party monitoring: NPR 1 Bn Anti-corruption unit: NPR 0.5 Bn Total: NPR 3 Bn
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Solution 8C: Research-to-Farm Pipeline

- NARC reform: mandate NARC to publish all research outputs in Nepali language extension materials within 12 months of completion — currently research sits in English journals unreachable to farmers
- NARC-Extension link: every NARC research station must host one 'Translation Officer' whose sole job is converting research findings into practical extension advice; 25 stations = 25 officers
- Competitive innovation fund: NPR 2 Bn fund for young Nepali agricultural researchers, entrepreneurs, and cooperatives to propose and test innovations — winner selected by open competition; successful innovations fast-tracked to national extension

BUDGET	NARC reform & Translation Officers: NPR 0.3 Bn Agricultural Innovation Fund: NPR 2 Bn Total: NPR 2.3 Bn
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PILLAR 9: ADDRESSING LABOUR EXODUS | NPR 22 BILLION

Problem Summary

Nepal loses 730,000–910,000 agricultural workers per year to international migration. The income gap is 3–10× between subsistence farming (NPR 30,000–50,000/year) and international wages (NPR 150,000–300,000/year). 32% of hill cultivated land lies fallow. Women and elderly are left to farm. Remittances fund construction, not agriculture. This problem cannot be solved without closing the income gap — which requires all other pillars to succeed first. But there are also direct interventions.

Solutions

Solution 9A: Make Agriculture Economically Competitive

The root cause is income. Pillars 1–6 directly address this. The specific additional interventions:

- High-value crop transition programme: subsidise transition from low-value cereals to high-value off-season vegetables, herbs, floriculture, and organic products in districts with high migration rates; a farmer on 0.5 ha of organic ginger can earn NPR 200,000+ — comparable to Gulf wages
- Agricultural entrepreneur development: select 1,000 young returnee migrants/year and provide them with: a NPR 500,000 low-interest agricultural enterprise loan (ACGF-backed), business training, and a market linkage to a guaranteed buyer — target those who saved money abroad and want to invest in Nepal
- Agri-tourism linkage: in hill districts, support combined farming + homestay agri-tourism (organic farms + tourist trekking stays) — diversifies income without requiring migration

BUDGET

High-value crop transition: NPR 8 Bn | Returnee entrepreneur programme: NPR 2 Bn (1,000/yr × NPR 500,000 each, partially recoverable as loans) | Agri-tourism: NPR 1 Bn | Total: NPR 11 Bn

Solution 9B: Labour Substitution Through Mechanisation

Where labour has already left, mechanisation is the only viable path to cultivation. Nepal must stop treating mechanisation as a luxury.

- Custom Hiring Centres (CHCs): government establishes 500 CHCs at local government level (concentrated in labour-scarce hill districts); each centre has: 2 tractors, transplanter, reaper, thresher, available for hire at subsidised rates
- Private CHC incentive: any private entrepreneur establishing a CHC with 10+ machines receives: 40% capital subsidy + 3-year GST exemption on machinery spare parts
- Drone technology for hill agriculture: pilot drone-based fertiliser and pesticide application in 5 high-altitude districts where terrain makes manual application extremely difficult; if successful, scale nationally
- Women-operated CHC priority: women's cooperatives operating CHCs receive an additional 10% capital grant — addresses gendered burden of labour exodus

BUDGET	500 government CHCs: NPR 5 Bn Private CHC subsidies: NPR 3 Bn Drone pilot programme: NPR 0.5 Bn Total: NPR 8.5 Bn
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Solution 9C: Remittance Channelling into Agriculture

- Agriculture Investment Bond: Finance Ministry issues NPR 200 Bn in 7-year agricultural bonds at 8% interest — above bank deposit rates; specifically marketed to the Nepali diaspora; proceeds fund Pillar 2 irrigation and Pillar 5 ACGF
- Matching fund: for every NPR 100 a non-resident Nepali invests in a registered agricultural cooperative, government contributes NPR 50 as a matching grant — catalyses diaspora agricultural investment
- Tax exemption on agricultural income: all income from agriculture by returning migrants for the first 5 years of return — reduces the financial disincentive to return

BUDGET	Bond marketing: NPR 0.2 Bn Matching fund: NPR 2 Bn Tax exemption (revenue forgone): estimated NPR 0.3 Bn/yr Total direct cost: NPR 2.5 Bn
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PART III: INTEGRATED FIVE-YEAR MASTER TIMELINE

Phase 1: Foundation (Year 1 — FY 2025/26) | NPR 55 Billion

Year 1 is about legislation, systems, and institutional foundations. Almost no physical construction — but the legal and institutional architecture must be built before capital can flow effectively.

Quarter	Legislative Actions	Institutional Setup	Programme Launch
Q1(Jul–Sep 2025)	Draft: Agricultural Functions Clarity Act, Groundwater Management Act, ACGF Act	National Agricultural Coordination Council formed; ACGF Board constituted	Land documentation mission deployed to all 77 districts; 6-source fertiliser supply negotiations
Q2(Oct–Dec 2025)	Parliament tables Agricultural Functions Clarity Act; Land documentation legal framework	NACC first quarterly meeting; Provincial Investment Plan guidelines issued; ADBL recapitalisation approved	Smart Farmer Card system designed and piloted in 5 districts; 100 LPP pilot cooperatives
Q3(Jan–Mar 2026)	Agricultural Functions Clarity Act passed; Groundwater Management Act in Parliament	7 Provincial Seed Farm sites selected and tenders issued; 77 District Aggregation Centre designs approved	JAEO recruitment: first 2,000 hired; NAKP app development contract awarded
Q4(Apr–Jun 2026)	ACGF Act passed; Seed Act amended (import duties abolished)	ACGF operational; Warehouse Receipt Regulations issued by NRB	50,000 farm ponds construction begun; Smart Farmer Card national rollout; 100 CHCs established

Phase 2: Infrastructure Build (Years 2–3 — FY 2026/27 to 2027/28) | NPR 140 Billion

The largest capital expenditure phase. Irrigation, market infrastructure, credit scaling, and extension deployment happen simultaneously.

Domain	Year 2 Targets	Year 3 Targets
Irrigation	200,000 ha new coverage; 150,000 farm ponds; 250 check dams; 500 WUAs operational	320,000 ha total; 200,000 ponds; all 7 provincial irrigation hubs under construction
Inputs	Smart cards covering 70% of farmers; 5 Provincial Seed Farms operational; 10 new climate varieties	Subsidy NPR 20 Bn; 500,000 farmers on improved seeds; domestic bio-fertiliser pilot producing
Market	50 District Aggregation Centres operational; cold chain subsidies deployed; NAKP launched	All 77 DACs operational; 7 Provincial Processing Hubs in construction; digital trade platform live
Credit	ACGF disbursing NPR 20 Bn private bank loans; warehouse receipt system operational	ACGF NPR 50 Bn cumulative; 35% of farmers with formal credit access

Extension	4,000 JAEOS deployed; 1 per 800 farmers; 100,000 farmers in Field Schools	6,000 JAEOS deployed; 1 per 500 farmers; Vocational Centres operational
Labour/Land	5,000 LPP cooperatives; 50,000 ha pooled; Land documentation 80%	Inheritance law enacted; 100,000 ha pooled; Land Bank operational; 300 private CHCs

Phase 3: Scale and Consolidation (Years 4–5 — FY 2028/29 to 2029/30) | NPR 145 Billion

Years 4–5 are about scaling what works, fixing what doesn't, and embedding sustainability. External aid dependence for core functions should end; government and private sector sustain the system.

Domain	Year 4 Targets	Year 5 Targets (2030)
Irrigation	430,000 ha; canal efficiency 50%+ nationally; WUA maintenance self-financing begins	500,000 ha new; 65% national coverage; conjunctive use standard practice in 3 provinces
Inputs	Subsidy NPR 16 Bn; 1 Mn farmers on improved seeds; fertiliser use 110 kg/ha avg	Subsidy NPR 14 Bn (sustainable); domestic bio-fertiliser 15% of supply; fertiliser use 140 kg/ha
Market	7 Processing Hubs operational; exports NPR 120 Bn; farmer price share 50%+	Exports NPR 180 Bn; post-harvest losses below 15%; farmers receive 50–60% of retail price
Credit	ACGF NPR 75 Bn cumulative; 40% farmers with formal credit; NPA rate < 8%	ACGF NPR 100 Bn; formal credit 50%+ for commercial farmers; ADBL financially self-sustaining
Extension	Extension ratio 1:500; 300,000 farmers/yr in Field Schools; NAKP 2 Mn users	Extension 1:450; Vocational Centres training 50,000/yr; all 1,100 graduates absorbed
Governance	100% subsidy through Smart Cards; NACC meetings regular; provincial plans executing	Consolidated 8 flagship programmes; 85%+ budget execution; no lapses; anti-corruption active

PART IV: MASTER BUDGET FRAMEWORK

Five-Year Investment Summary by Pillar

Pillar	Problem Addressed	Y1 (NPR Bn)	Y2 (NPR Bn)	Y3 (NPR Bn)	Y4 (NPR Bn)	Y5 (NPR Bn)	Total(NPR Bn)
1. Land	Fragmentation, documentation	3	4	4	4	3	18
2. Irrigation	Coverage, storage, groundwater	8	20	22	18	17	85
3. Inputs	Fertiliser, seeds (excl. subsidy)	3	8	9	8	6	34
4. Markets	Infrastructure, exports, digital	2	6	8	7	5	28
5. Finance	ACGF, ADBL, credit reform	14	7	5	5	4	35
6. Extension	JAEOs, curriculum, digital	3	5	5	3	2	18
7. Climate+Soil	Resilience, soil health	2	5	7	7	5	26
8. Governance	Institutions, coordination	1	1	1	1	1	5 (direct)
9. Labour	Mechanisation, returnees	3	5	6	5	3	22
Fertiliser Subsidy (Reformed)	Smart targeted subsidy	26	20	20	17	14	97
TOTAL		65	81	87	75	60	368
vs. Current Budget Trajectory		57.5	58	59	60	61	295.5
INCREMENTAL INVESTMENT		+7.5	+23	+28	+15	-1	+72.5

The incremental investment of NPR 72.5 Bn over 5 years beyond current trajectory is largely self-financing: reformed fertiliser subsidies (saving NPR 70 Bn over 5 years versus current trajectory) and ACGF leveraging NPR 100 Bn in private lending fund most of the expansion.

Funding Sources

Source	Mechanism	Estimated 5-Year Contribution
Government budget reallocation	Redirect fertiliser subsidy savings to capital investment	NPR 70 Bn
Federal budget increment	5–8% annual increase in agriculture allocation within Investment Decade commitment	NPR 30 Bn
World Bank / ADB irrigation loans	Concessional loans for large irrigation infrastructure (3% interest, 25-yr term)	NPR 60 Bn
ACGF private bank leverage	Government NPR 20 Bn seed capital unlocks NPR 100 Bn private lending (off-budget)	NPR 100 Bn (private)
Agricultural Investment Bonds (diaspora)	7-year bonds at 8% marketed to non-resident Nepalis	NPR 50 Bn
Provincial + local government contribution	Restored sub-federal agriculture funding (15% of MoALD budget)	NPR 30 Bn
Private sector co-investment	Processing hubs, cold chain, CHCs under government incentive framework	NPR 25 Bn

PART V: RISK REGISTER AND ACCOUNTABILITY FRAMEWORK

5.1 Critical Risks

Risk	Likelihood	Impact	Mitigation
Political instability interrupts long-term projects	High	High	All 8 flagship programmes given constitutional protection; NACC approval required to cancel; opposition parties signed onto cross-party Agricultural Compact
Fertiliser subsidy reform creates farmer backlash	High	Medium	Smart card gives SAME total subsidy to poor farmers — only elite capture is cut; extensive farmer communication campaign; pilot in 3 districts before national rollout
Irrigation capital budget not executed (historical pattern)	High	High	Mandate single project management unit per province; simplified procurement for schemes under NPR 50 Mn; quarterly execution dashboards published
ACGF becomes non-performing loan repository	Medium	High	Strict qualifying criteria; 70% guarantee (not 100%) keeps bank risk; agricultural insurance (Pillar 7) reduces default cascade; quarterly portfolio review
Climate shock disrupts Year 2–3 infrastructure investment	Medium	High	All designs must meet climate risk standards from district mapping; insurance programme protects farmers; 15% contingency budget in each pillar
India fertiliser MOU disruption (already occurred)	High	High	Immediate: sign supply agreements with China, Middle East; 90-day strategic reserve maintained permanently; accelerate domestic bio-fertiliser production
Extension officer deployment fails due to urban preference	Medium	Medium	Home-district deployment rule; performance bonus tied to local yield improvements; motorbike and digital tools make rural posting attractive

5.2 Accountability Architecture

Quarterly Performance Dashboard (Public)

The Agriculture Minister personally presents a quarterly performance dashboard to Parliament and publishes it online within 15 days of each quarter end. The dashboard covers:

- Budget execution rate by pillar (target: 85%+ by Q3 each year)
- Physical targets: hectares irrigated, JAEs deployed, farm ponds constructed, cooperatives registered, loans disbursed

- Outcome indicators: average yield by crop and province, fertiliser use rate, credit access rate, export value
- Red-flag reporting: any pillar below 70% execution automatically triggers a corrective action review within 30 days

Independent Mid-Term Evaluation (Year 3)

A fully independent panel (international agricultural economists + Nepali civil society) conducts a comprehensive mid-term evaluation in Year 3. Findings are public. Ministry must respond formally within 60 days. Parliament debates the response.

Farmer Feedback Mechanism

Every farmer in any government programme receives a QR code linking to a simple 5-question satisfaction survey. Results aggregated monthly and reviewed by NACC. Consistently low-rated programmes face restructuring. This is non-negotiable — farmer voice must drive programme evolution.

PART VI: 100-DAY QUICK WINS FOR IMMEDIATE ACTION

These actions require no new legislation, no new budget appropriation, and can begin within 100 days of this roadmap's adoption:

Action	Who Is Responsible	Timeline	Expected Impact
Sign emergency fertiliser supply MOU with China and Middle East suppliers to replace expired India MOU; establish 30-day buffer stock immediately	Agriculture Minister + Finance Minister + Foreign Minister	Day 1–30	Eliminate fertiliser supply crisis for Kharif 2026 season
Issue NRB directive to all A-class banks: comply with 5% agricultural lending mandate or face 0.5% fine on total lending	Nepal Rastra Bank Governor + Finance Ministry	Day 1–15	Unlock NPR 15–20 Bn in private bank agricultural lending immediately
Freeze all new fertiliser subsidy disbursement to non-farmers (traders, non-registered parties); register all subsidy recipients biometrically	MoALD Secretary	Day 1–30	Redirect NPR 3–5 Bn/yr currently lost to elite capture
Order all 77 District Agriculture Knowledge Centres to open a climate emergency advisory desk — immediate guidance on flood-tolerant varieties for upcoming season	Department of Agriculture Director-General	Day 1–15	Preparedness for 2026 monsoon season
Announce the Land Pooling Programme publicly: terms, grants, eligibility. Call for applications from cooperatives to be pilot participants	Agriculture Minister (press announcement)	Day 30	Signal to farmers and cooperatives that change is coming; begin Year 1 pipeline
Commission the district agricultural climate risk maps from DHM — simple assignment requiring no budget allocation, only inter-ministry order	Agriculture + Water Resources Ministries joint directive	Day 1–7	Data for Year 1 climate-adaptive variety allocation
Publish the consolidated 8-Flagship Programme list: abolish redundant budget line items; single accountable officer for each flagship	MoALD + Ministry of Finance joint order	Day 30–60	Eliminate budget fragmentation; create accountability lines
Recruit and deploy 500 JAEs immediately from existing civil service reserve lists — no new hiring process needed for first cohort	Public Service Commission coordination	Day 30–60	Begin closing extension gap before new hiring cycle

CONCLUSION: THE CHOICE BEFORE PARLIAMENT

Nepal's agricultural crisis is not a mystery. The root causes are documented, the solutions are known, and the budget exists — it is simply allocated to the wrong things.

The choice before lawmakers and the Agriculture Minister is stark:

If Nepal Continues Current Path	If Nepal Adopts This Roadmap
NPR 28 Bn/yr on fertiliser subsidies reaching 30–40% of farmers	NPR 14 Bn targeted subsidy reaching 90%+ via Smart Cards — saving NPR 14 Bn for investment
36% irrigation coverage; 50% yield potential; monsoon-trapped agriculture	65% irrigation coverage; 3× yield potential; year-round farming for 2 million households
730,000 farmers leaving agriculture each year	Commercial farming income approaches international wage levels in high-value crops
1 extension worker per 1,500 farmers; textbook advice from Kathmandu	1 per 500; locally-deployed; performance-incentivised; farmer-problem driven
Farmers receive 30% of retail price; NPR 60 Bn exports	Farmers receive 50–60% of retail price; NPR 180+ Bn exports
6.7% of credit in pre-harvest period; informal moneylenders at 36–60%	35%+ of farmers with formal credit at 10–12%; pre-harvest capital accessible
32% of hill land fallow; food insecurity growing	Hill land revitalised; national food self-sufficiency restored
~1% agricultural growth rate; agriculture draining fiscal resources	4–5% agricultural growth; agriculture generating fiscal surplus and export income

The total incremental cost of transformation is NPR 72.5 billion over 5 years — largely self-funded through subsidy reform. The cost of inaction is the continued haemorrhaging of Nepal's rural population, the permanent loss of hill agricultural land, and a structural food and fiscal crisis by 2030.

This roadmap does not ask for charity. It asks for structural reform: redirect existing money, fix broken institutions, invest in infrastructure with lasting returns, and hold every official accountable for results. Nepal has declared an Agriculture Investment Decade. This roadmap is the operational plan to make that declaration mean something.